Comparison between different numerical methods for the FPUT problem

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I. Introduction

The FPUT problem is a numerical experiment carried out for the first time in 1953, just one year before Fermi's death. It consist in simulating a system of N particles interacting through an harmonic potential plus a non linear perturbation. The boundary condition impose that the first and the last springs are fixed to a wall. The problem was designed by Fermi in order to check the idea that essentially any non-linearity would lead to a system satisfying the ergodic hypothesis and to estimate the time of thermalization of the system, namely the time required to reach equipartition of energy. The outcome was quite against Fermi's conjecture and a new era was opened to understand why ergodicity was violated. It is mandatory to cite that the numerical experiment was carried out on the first ever computer, the MANIAC I by Mary Tsingou.

II. Hamilton's equations

The hamiltonian of the system is given by:

$$\mathcal{H}(q,p) = \sum_{i=1}^{N} \left[\frac{p_i^2}{2m} + \frac{k}{2}(q_{i+1}-q_i)^2 + \frac{\alpha}{3}(q_{i+1}-q_i)^3 \right] \quad (1)$$

where M is the mass of the atoms, p_i is the momentum, k is the harmonic elastic coefficient, α is the coefficient of the cubic anharmonic term [1], and q_i is the displacement of the i-th atom from its equilibrium position. The solution to the corresponding linear problem $\alpha = 0$ is a periodic vibration of the string. If the initial position of the string is, say, a single sine wave, the string will oscillate in this mode indefinitely. Turning on the non-linearity $\alpha \neq 0$ the string would assume more and more complicated shapes, and, for t tending to infinity, would get into states where all the different modes acquire increasing importance. For this reason, Fermi and his collegues decided to plot the trend of modes' energy over time. The astonishing original results is shown in figure 1. There are no better words to describe what you're seeing than what the authors comment in the original article *We observe initially a gradual increase of energy in the higher modes as predicted. Mode 2 start increasing first, followed by mode 3, and so on. Later on, however this gradual sharing of energy among successive modes ceases. Finally, at a later time mode 1 comes back to within one percent of its initial value so that the system seems to be almost periodic.*[2]

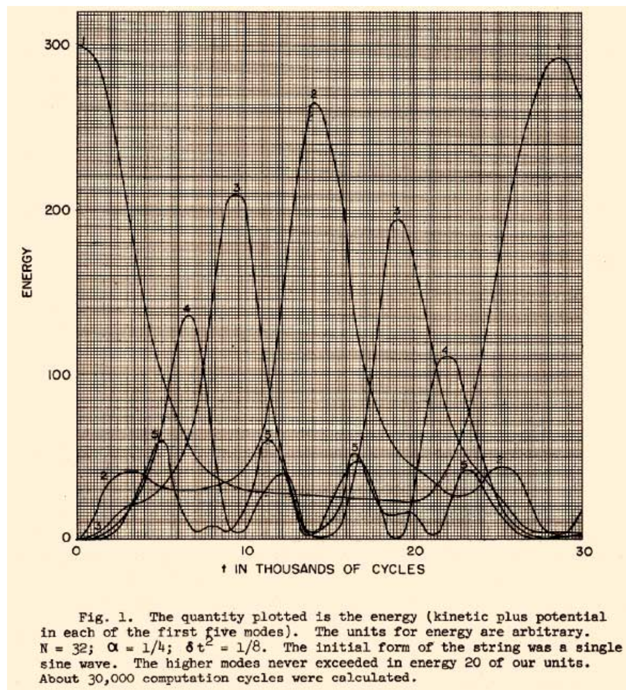

Fig. 1. The evolution of normal mode energy in Fermi original paper.

The dynamics of the examined physical system can be described starting from the Hamiltonian (1) and the corresponding Hamiltonian equations. The latter consider the partitionable expression of H.

$$H(q,p) = T(p) + U(q) \quad (2)$$

It reduces to a system of $2N$ ordinary differential equations of the form:

$$\begin{cases} \dot{q}_i = \frac{\partial T}{\partial p_i} = \frac{p_i}{m} \\ \dot{p}_i = -\frac{\partial U}{\partial q_i} = k(q_{i+1} - 2q_i + q_{i-1}) + \\ \qquad +\alpha[(q_{i+1}-q_i)^2 - (q_i - q_{i-1})^2] \end{cases} \quad (3)$$

The chain of harmonic oscillators described by the problem, being an isolated system, conserves total energy, which be-

[1] In the original work, a fourth-order nonlinear term was also considered. In this work, we have restricted ourselves to the case of cubic potential

[2] In this simulation energy is initially placed all in the first mode.

comes a crucial indicator for the quality of numerical methods in the absence of an analytical solution, considering the system's atypical statistical properties. Therefore, our work focused on the idea of initially considering the methods presented in the course, and then moving on to the exploration of similar methods but more suitable, especially in ensuring energy conservation: the so-called *symplectic methods*.

There are some common feature that we decided to adopt globally in our work:

- Number of particle is equal to 32
- Initial energy is placed only in the first mode
- Initial velocities are null
- Mass are all equal and unitary
- $\frac{\alpha}{\kappa} = 0.1$

The first part of this work, motivated by our intrest, was to reproduce the significant results of the original work. We let the reader the curiosity to check whether our codes faithfully reproduce the results of figure 1. Before moving to the presentation of the different methods implemented there is still an important physical aspect that has engaged the best minds of the last century. This has to do with one of the main feature of this problem: equipartition of energy. According to equipartition principle, one should observe that, after a certain amount of time, called time of thermalization the mean energy per mode approach the value: $\bar{E_k} = \frac{E_{tot}}{N}$. Nevertheless if you observe figure 2, where the average energy in mode k is plotted as a function of observation time T,

$$E_k(T) = \frac{1}{T}\int_0^T E_k(t)\,dt, \quad with\ k = 1, \ldots, N$$

it is clear that this fundamental principle of statistical physics is violated. The results obtained in the Fermi-Pasta-Ulam (FPU) experiment were decidedly contrary to what was expected, and Fermi himself, as Ulam wrote, was surprised and expressed the opinion that an important discovery had been made. This discovery unequivocally demonstrated how prevailing beliefs about the generality of mixing and thermalization properties in nonlinear systems might not always be justified. Fermi had been invited to give a prestigious lecture (the Gibbs Lecture) at the 1955 American Mathematical Society conference, and he had decided to speak specifically about the results of the FPU experiment. Due to the illness that led to his death, he never delivered the lecture. The solution to the problem, which constitutes an important chapter in modern mathematical physics, had paradoxically already been found in 1954, a year before the drafting of the Los Alamos note and unknownst to the authors, by the Soviet mathematician A. N. Kolmogorov. Without entering in the complicated details of the so called KAM theorem, which by the way is not the purpose of this work, we limit to discuss the main consequence that descend directly form the theorem. For a fixed number of particles N, the scenario is essentially as follows: at a given energy density $\epsilon = \frac{E}{N}$, there exists a threshold ϵ_c for the intensity of the perturbation, such that

- if $\epsilon < \epsilon_c$, the system does not equipartition

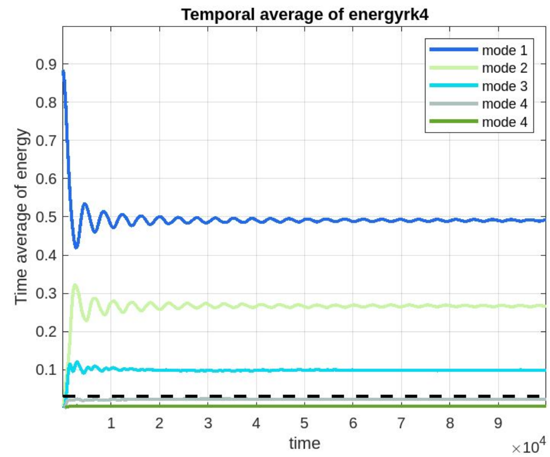

Fig. 2. Average energy over time for different modes.

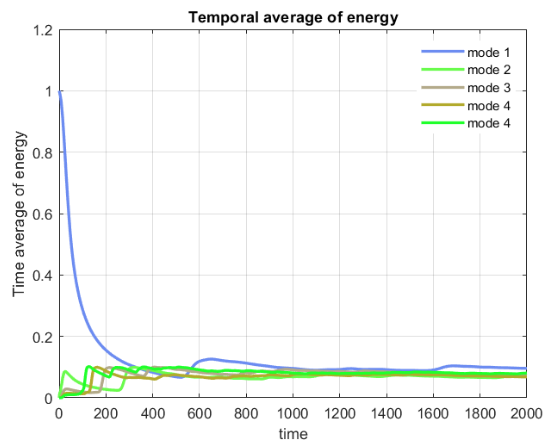

Fig. 3. Average energy over time with initial energy equal to 40. Step size of 0.01 and $\sqrt{\frac{k}{M}} = 30$. This time, equipartition of energy is reached

- if $\epsilon > \epsilon_c$, the system equipartitions, and there is agreement with the usual statistical mechanics.

It is easy to convince oneself that, with greater relevance to physical situations, fixing the value of the perturbation α, the energy density takes on the role of a control parameter, and there exists a threshold value ϵ_c that separates regular behavior from chaotic behavior. Since in our simulation N is fixed, the only way to hopefully observe the equipartition of energy is to change the initial energy. in figure 3 one can appreciate that equipartition in reached.

III. General approach to the problem adopted

Since we are dealing with a multi-body vibrational problem is quite natural to consider the normal modes:

$$a_k = \sqrt{\frac{2}{N+1}}\sum_{i=1}^{N} x_i \sin\left(\frac{ik\pi}{N+1}\right), \quad k = 1, \ldots, N \tag{4}$$

such that the system is reduced to N non-interacting harmonic oscillators with angular frequencies given by

$$\omega_k = 2\sqrt{\frac{K}{m}} \sin\left(\frac{k\pi}{2(N+1)}\right) \quad k = 1, \dots, N$$

and energies given by

$$E_k = \frac{1}{2}\left(\dot{a_k}^2 + \omega_k^2 a_k^2\right) \tag{5}$$

Since we decided to put the energy in the first mode we started our code, calculating normal modes with Eq. 4. Indeed we proceed by defining a matrix A where

$$(A)_{i,k} = \sqrt{\frac{2}{N+1}} \sin\left(\frac{ik\pi}{N+1}\right)$$

where i and k goes from 1 to N. Now if we define the vector x_i of displacement from equilibrium positions we can compute the initial position of atoms solving the linear system:

$$\bar{x}_{t=0} = \mathbf{A}^{-1} \cdot \bar{a}_k$$

where from Eq. (5) we get $\bar{a}_{k=1} = \frac{\sqrt{2E_k}}{\omega_k}$ and $\bar{a}_{k\neq 1} = 0$ simply because $E_k = 0 \quad \forall k \neq 1$. Once we get the vector of initial positions we give it to the solver which will compute the new displacement for each particle at the next time step. Once is reached the final time, we will know each displacement in each time instant. We can therefore compute $\mathbf{A} \cdot \bar{x} = \bar{a}_k$ to get the normal modes, calculate the energy for each mode with Eq. 5 and eventually plot it. Indeed we decided to adopt the formalism of matrixes during all the work carried out. In particular when considering the system of first order ODE one have to solve for each particle:

$$\begin{cases} \dot{x}_n = v_n \\ \dot{v}_n = \kappa(x_{n+1} - 2x_n + x_{n-1}) + \\ \quad \alpha((x_{n+1} - x_n)^2 - (x_n - x_{n-1})^2) \end{cases} \tag{6}$$

Where n denotes the particle index. We can defined four matrix $N \times N$ such that:

$$\begin{cases} \dot{\bar{x}} = \mathbf{I} \cdot \bar{v} \\ \dot{\bar{v}} = \kappa \mathbf{L} \cdot \bar{x} + \alpha((\mathbf{P} \cdot \bar{x})^2 + (\mathbf{N} \cdot \bar{x})^2) \end{cases} \tag{7}$$

where $\mathbf{I}$ is a diagonal matrix while $\mathbf{L}$, $\mathbf{P}$, $\mathbf{N}$ have the form:

$$\mathtt{L} = \begin{bmatrix} -2 & 1 & 0 & \cdots & 0 \\ 1 & -2 & 1 & \cdots & 0 \\ 0 & 1 & -2 & \ddots & \vdots \\ \vdots & \vdots & \ddots & -2 & 1 \\ 0 & 0 & \cdots & 1 & -2 \end{bmatrix}$$

$$\mathtt{P} = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & \cdots & 0 \\ 0 & 0 & -1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & -1 & 1 \\ 0 & 0 & \cdots & 0 & -1 \end{bmatrix}$$

and $\mathbf{N}$ is essentially equal to $\mathbf{P}$ with the only difference that the superior diagonal is null and the inferior diagonal is full of ones. Now the reader could ask why do we need to express the problem in such a way. Well, firstly it is indubitable more elegant, not only from a mathematical point of view but also regarding the coding part. Secondly, it can improve the performance of Matlab especially if one want to go very high with the number of particles, which by the way was not in our interests.

IV. Methods for Hamiltonian problem

A. Introduction

Our work began by applying the methods introduced in the course [2] and their extension to first-order ODE systems [1]. The initial objective was to qualitatively reproduce the dynamics depicted in Figure 1. Both the *Explicit Euler* and *Runge-Kutta* methods were found unsuitable for this task. The inadequacy of the Explicit Euler method was expected, given its stability region that excludes the imaginary axis—where, neglecting perturbations, the eigenvalues of our system lie. The Runge-Kutta method also failed even under favourable numerical conditions.

A different approach was then taken, considering the specific structure of the equations. Notably, the derivatives of positions $\dot{q}$ were observed to depend solely on velocities $v = \frac{p}{m}$, and vice versa. As an empirical solution, a numerical scheme was adopted, where at each time instant $t+1$:

1) $v(t+1)$ is explicitly computed using $q(t)$.
2) $q(t+1)$ is then calculated based on the recently determined $v(t+1)$.

Remarkably, this approach yielded good qualitative results and is implemented in the codes `EulerMethod.m` and `rk4.m` in the `Methods` folder. However, before delving into more detailed analyses, we decided to take a step back and explore the theory of symplectic Runge-Kutta methods [5].

B. Runge-Kutta methods

The general form of an $s-stage$ Runge-Kutta for solving an ordinary differential equation is given by:

$$\begin{gathered} k_i = f(t_n + c_i h, y_n + h\sum_{j=1}^{s} a_{ij} k_j) \quad i = 1, \dots, s \\ y_{n+1} = y_n + h\sum_{i=1}^{s} b_i k_i \end{gathered} \tag{8}$$

where:

- h is the step size,
- t_n and y_n are the current time and solution values,
- $f(t, y)$ is the differential equation,
- c_i, a_{ij}, and b_i are real coefficients that fully define the method.

The coefficients, and so the method, can be describe through a so called Butcher Tableau:

$$
\begin{array}{c|cccc}
c_1 & a_{11} & a_{12} & \cdots & a_{1s} \\
c_2 & a_{21} & a_{22} & \cdots & a_{2s} \\
\vdots & \vdots & \vdots & \ddots & \vdots \\
c_s & a_{s1} & a_{s2} & \cdots & a_{ss} \\
\hline
 & b_1 & b_2 & \cdots & b_s
\end{array} \tag{9}
$$

Depending on the structure of the matrix one can distinguish:

- Explicit Runge-Kutta Method (ERK): $a_{ij} = 0$ for $i \leq j$.
- Implicit Runge-Kutta Method (IRK) : at least one $a_{ij} \neq 0$ for $i < j$.
- Diagonal implicit Runge-Kutta Method (DIRK): $a_{ij} = 0$ for $i < j$ and at least one $a_{ii} \neq 0$.

We will not consider the case of implicit methods, generally more complex, as they require the solution of systems of nonlinear equations. Before discussing how the other two types serve the problem at hand, for completeness, we state the following theorem:

Theorem 1. *An explicit Runge-Kutta method with s stages cannot have an order greater than s. Additionally, there are no Runge-Kutta methods with s stages and order s if $s \geq 5$.*

C. Partitioned Runge-Kutta method

Considering our system,it is described by a separable Hamiltonians that as said before bring to a system of equation like 3 where there are two kind of equation one for momenta (or better speeds) and one for positions. In this situation, an help can come looking for *partitioned Runge-Kutta methods*, which basically consist on implementing two different RK methods for position and speed equation, namely it means having two Butcher Tableau. Considering the particular forms of our equation the two methods can be state in slightly different way with respect of 8.

1) First we introduce two different increment at each step: k_i for speeds v and l_i for positions x [3]. So that:

$$
v_{n+1} = v_n + h \sum_{i=1}^{s} b_i k_i \quad x_{n+1} = x_n + h \sum_{i=1}^{s} \hat{b}_i l_i \tag{10}
$$

2) It is convenient to introduce two terms, g_i for speeds and f_i for positions, in order evaluate at each step the dependency from other one, at time t_n:

$$
g_i = v_n + h \sum_{j=1}^{s} a_{ij} k_i \quad f_i = x_n + h \sum_{j=1}^{s} \hat{a}_{ij} l_i \tag{11}
$$

3) Now recalling the two type of differential equation and their variable dependence 3, one can express:

$$
k_i = -\frac{\partial \mathcal{H}}{\partial x}(f_i, g_i) = -\frac{\partial U}{\partial x}(f_i) \quad l_i = \frac{\partial \mathcal{H}}{\partial p}(f_i, g_i) = g_i \tag{12}
$$

Finally these coefficient can be use to evaluate 10 solutions.

From this point of view the method can be represented with two Butcher Tableau, that in the case of symplectic methods assume a very particular form, as we should see in next section.

[3]In our case they are vectors of N elements

D. Symplectic Runge-Kutta methods

Despite this methods have precise mathematical and geometric meaning and property, we won't intricately address this details. For our purposes, we recognise them as potent tools for Hamiltonian systems, renowned for their stability and energy conservation in numerical simulations, that in general make them good tools for instance in fields like Quantum Mechanics and astronomy.

Coming to Runge-Kutta methods a very powerful theorem can be state: [5]:

Theorem 2. *If the $s \times s$ matrix M with elements*

$$
m_{ij} = b_i a_{ij} + b_j a_{ji} - b_i b_j, \quad i, j = 1, \ldots, s
$$

satisfies $M = 0$, then the Runge-Kutta method is symplectic.

With the following significant consequence:

Theorem 3. *Explicit Runge-Kutta methods are never symplectic.*

Moving to the case of *Partitioned Runge-Kutta* with a separable Hamiltonian there is the following , more concise condition on method's coefficients:

Theorem 4. *If the coefficients of partitioned method satisfy*

$$
\hat{b}_i a_{ij} + \hat{b}_j a_{ji} - \hat{b}_i \hat{b}_j = 0, \quad i, j = 1, \ldots, s \tag{13}
$$

Then the method is symplectic.

A this point one can choose a method satisfying:

$$
a_{ij} = \begin{cases} 0 & \text{for } i < j \quad \text{diagonally implicit} \\ \leq 0 & \text{for } i \leq j \quad \text{explicit.} \end{cases} \tag{14}
$$

Since $\frac{\partial H}{\partial x}$ depends only on x, the method is in practice explicit since for a certain k_i does not depend on itself but on l_i . Whit this assumption, the symplecticity condition of theorem 4 imply:

$$
a_{ij} = b_j \text{ for } i \geq j, \quad \hat{a}_{ij} = \hat{b}_j \text{ for } i > j. \tag{15}
$$

The method is so characterised by the two matrix:

$$
\begin{array}{c|cccc||cccc}
 & b_1 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\
 & b_1 & b_2 & \cdots & 0 & \hat{b}_1 & 0 & \cdots & 0 \\
 & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\
 & b_1 & b_2 & \cdots & b_s & \hat{b}_1 & \hat{b}_2 & \cdots \hat{b}_{s-1} & 0 \\
\hline
 & b_1 & b_2 & \cdots & b_s & \hat{b}_1 & \hat{b}_2 & \cdots \hat{b}_{s-1} & \hat{b}_s
\end{array} \tag{16}
$$

That can be summarised just considering vectors:

$$
\begin{array}{ll}
\mathbf{b}: & b_1, b_2, \cdots, b_s \\
\hat{\mathbf{b}}: & \hat{b}_1, \hat{b}_2, \cdots, \hat{b}_s
\end{array} \tag{17}
$$

E. Euler method

we noticed that our initial working but naive idea , about evaluate first speeds explicitly and then using this result to evaluate positions, had so a theoretical motivation, at least for Euler method. That as we know in general it can be seen as a first order RK.
In this case choosing in the previous scheme a combination of the implicit Euler method $b_1 = 1$ with the explicit Euler method $\hat{b}_1 = 1$ gives the symplectic method of order 1:

$$p_1 = p_0 - h\frac{\partial U}{\partial q}(q_0) \quad q_1 = q_0 + h\frac{\partial T}{\partial p}(p_1). \tag{18}$$

That is exactly the one that we already had in our code.

F. Runge Kutta 4 and modified version

Notably ERK 4 in complete accordance with theorem 3 does not conserve energy, on the contrary, it is practically always unstable for the periods of interest in the analysis. Considering, as explained in this chapter introduction, the value of speed at $t+1$ to evaluate $x(t+1)$, we have obtained at least qualitative results. However moving to a more quantitative analysis we have observed a very limited stability region, an empirical order of convergence below 1 and more energy fluctuations around the exact value. This must not surprise since one can check that the coefficients of such an artisanal method, does not satisfy the condition of theorem 2. For that reason in the numerical analysis of next chapter we are not going to consider it.

G. Ruth Method

A possible third order scheme with s = 3 was found by Ruth in 1983. We have implemented it for our problem considering the steps mentioned before and the scheme:

$$\begin{array}{cccc} \mathbf{b}: & \frac{7}{24} & \frac{3}{4} & -\frac{1}{24} \\ \hat{\mathbf{b}}: & \frac{2}{3} & -\frac{2}{3} & 1 \end{array} \tag{19}$$

H. Leapfrog Method

Despite the elegance of Hamiltonian equations, one could try to solve directly the equation of Newton. This means that we'll solve N equations instead of 2N but obviously there is a price: now the equation are of second order in the form:

$$\ddot{x}_n = \kappa(x_{n+1}-2x_n+x_{n-1})+\alpha((x_{n+1}-x_n)^2-(x_n-x_{n-1})^2) \tag{20}$$

with the usual initial condition already discussed. The method is obtained by simply applying the finite difference formula for the discretization of the second derivative:

$$\delta^2 x_i \approx \frac{x_{t+1} - 2x_t + x_{t-1}}{h^2}$$

It is clearly an explicit method and can be written as

$$x_{t+1,n} = 2x_{t,n} - x_{t-1,n} + h^2 a(x_t, t) \quad t = 1, ..., T_{max}$$

Finally, it is possible to demonstrate that the method is second order consistent.

I. Symplectic Nyström Method

R.H. Merson: "*...I have not seen the paper by Nyström. Was it in English?*"
J.M Bennett: "*In German actually, not Finnish*"

Consider the problem

$$\dot{p} = -\frac{\partial U}{\partial q}(q) \qquad \dot{q} = Mp \tag{21}$$

where M is a symmetric matrix (in our case is also diagonal). Nyström methods for the system are given by

$$Q_i = q_o + c_i hM\dot{p}_o + h^2\sum_j^s \bar{a}_{ij} M l_j \quad l_j = -\frac{\partial U}{\partial q}(Q_j)$$

$$q_1 = q_0 + hM\dot{p}_o + h^2\sum_i \bar{b}_i M l_i \quad p_1 = p_o + h\sum_i^s b_i l_i$$

Theorem 5. *(Suris 1989) Consider the system (21). Then the s-stage Nyström method is symplectic if the following two conditions are satisfied:*

$$\bar{b_i} = b_i(1 - c_i)$$

$$b_i(\bar{b_j} - \bar{a}_{ij}) = b_j(\bar{b}_i - \bar{a}_{ji})$$

with i,j = 1,...,s

For Nyström methods satysfing both conditions one can assume without loss of generality that

$$b_i \neq 0 \quad \text{for} \quad i = 1, \ldots, s$$

Under this assumption the condition given by the theorem 5 implies that:

$$\bar{a}_{ij} = b_j(c_i - c_j) \quad \text{for} \quad i \geq j$$

while

$$\bar{a}_{ij} = 0 \quad \text{for} \quad i \leq j$$

The particular form of the coefficient $\bar{a}_{ij}$ allows the following simple implementation.

$Q_o = q_o \quad P_o = p_o$
for $i := 1$ to **s** do
$Q_i = Q_{i-1} + h(c_i - c_{i-1})MP_{i-1}$
$P_i = P_{i-1} - hb_i\frac{\partial U}{\partial q}(Q_i)$
$q_1 = Q_s + h(1 - c_s)MP_s, \quad p_1 = P_s$ with $c_0 = 0$

In our code, we used a Nyström method of 3-stages with coefficients: $c_1 = 1 - c_3 = \frac{1}{6}(2 + \sqrt[3]{2} + \frac{1}{\sqrt[3]{2}})$ and $c_2 = \frac{1}{2}$

V. NUMERICAL RESULTS

A. Stability

Convergence is a fundamental concept; an algorithm that does not converge would lack practical significance. However, it's important to note that convergence alone does not necessarily guarantee that a method will yield acceptable numerical results. It becomes imperative to ensure that the numerical solution possesses certain properties akin to the exact solution. While the convergence of an algorithm signifies that it approaches a solution, it doesn't inherently imply accuracy or reliability. Indeed stability and convergence are two strictly linked arguments but it's not the case to go deeply into this vast topic. Therefore what follows will be just qualitatively. We know that the system describing an harmonic oscillator has imaginary eigenvalues and that each of the methods described above has its own stability region. It is essential, in order to reach an appropriate solution, to make sure that we are inside this stability region. Knowing whether we are inside the region of stability or not is essential especially when time simulation became very long and therefore one want to know in advance if the method will lead to a reliable solution. Now, comes the crucial question, what are the parameter responsible for the stability of the system? It turns out that both h, namely the time step and ω_{max}, the maximum frequency of oscillation are the crucial parameter to take under control, in order to avoid useless running of methods that explodes into nonphysical solutions. In particular the condition that has to be satisfied has the form of $\omega \cdot h \leq c$ where c is a number that depends on the particular method adopted. Since we weren't fully aware of the stability region of each method we tried in a brutal and surely not elegant way. We started choosing one particular value of $\sqrt{\frac{K}{M}}$ which will automatically fix $\omega_{max} = 2\sqrt{\frac{k}{M}} sin(\frac{N\pi}{N+1})$. It's clear that the maximum frequency will correspond to the lowest period of oscillation. It should be evident that the step size chosen is a crucial parameter in order to correctly describe the mode with the lowest period. For instance, if the step size is bigger than the smaller period of oscillation, this mode will never be correctly treated and will eventually bring the methods to blow up. Once we set a particular value for ω_{max} we started to run the code with different step size in order to identify the border line between stability and instability. What we have found is that for Ruth of order three the constant c takes the value of 2.4 while for Nyström of order three we found $c = 1.7$. For the other methods the border line between stability and instability region was more difficult to deline in the sense that for some value of h and ω the system didn't explode but still presented lots of oscillations. It' s mandatory to stress that these considerations does not constitute a rigorous results but just a qualitatively observations that we made along our route.

An important quantity that we decided to investigate was the trend of the error over total energy varying T_{max}, namely the total time of simulation and leaving all other parameter unchanged. What we have found is shown in fig.4 where you can appreciate the difference between the methods implemented.

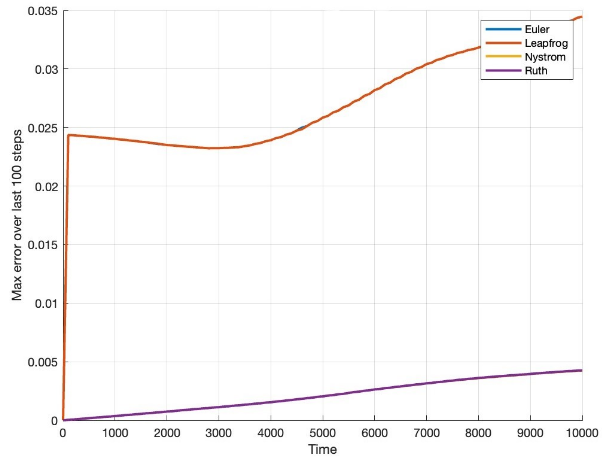

Fig. 4. This figure shows the maximum error on the total energy when varying the total time of simultaion

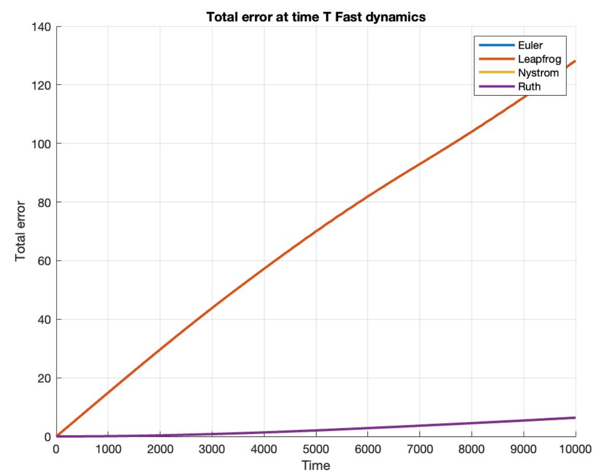

Fig. 5. Trend of total error over time for a fast dynamic.

It is clear that in the Routh and Nystrom methods the error remain bonded and its rate of growth is significantly less than the Leapfrog and Euler method[4]. However observe that even if the Leapfrog method present a rapid growth for small value of T_{max} then it continue to increase less quickly and it reaches the modest value of 0.035 for the final time $T_{max} = 10000$. Therefore, Routh and Nystrom seems to conserve the energy definitely better than Leapfrog and Euler but also the the latter aren't that bad. To show and underlie the supremacy of the Routh and Nystrom methods over the others we decided to plot the total error accumulated over time. The results are shown in figure 5 where one can appreciate how the total error grows over time but with significantly different rate between the methods.

[4] In these plots the Routh methos covers the line of the Nystrom methods and the Leapfrog's line covers that one of Euler

B. Order of convergence

From [2] and [1] we recall the two following definitions that are meaningful for the following:

Definition 1 (Order of convergence)**.** *If it can be demonstrated that, for a given quantity α, the approximation $\alpha_{approx}(h)$ satisfies the error estimate*

$$\alpha_{approx}(h) = \alpha + ch^p + o(h^p)$$

for sufficiently small h, then $\alpha_{approx}(h)$ is termed an approximation of order p for α. More precisely, one would state that $\alpha_{approx}(h)$ converges of order p as h approaches 0.

In order to validate the correctness of our implementations and understand the behaviour of the numerical methods within the context of this problem in comparison to theoretical expectations, we aimed to provide an empirical estimation of the convergence order. The following definition proves useful for this purpose:

Definition 2 (Empirical Estimate of Convergence Order)**.** *Given a numerical method $\alpha_{approx}(h)$ depending on the parameter h for the approximation of α, the empirical estimate of the convergence order is given by the number*

$$p_{emp} = -\log_2\left(\frac{|\alpha_{approx}(h/2) - \alpha|}{|\alpha_{approx}(h) - \alpha|}\right). \tag{22}$$

In our case, there were several possibilities to evaluate p_{emp}. One option was to use the only exact data available in the absence of an analytical solution, namely the value of the total energy. However, this would be assessed later as it is useful but does not allow for a comparison with theory, which concerns the error on the solution. The other approach was to resort to the `ode45` solver in Matlab, configured with rather stringent parameters such as `RelTol` $= 10^{-10}$ and `MaxStep` set to the numerical step under consideration at that moment.
At this point, we developed the code in the `error` folder. In particular, the `test_ode` script allows solving the problem for a specific value of K and iterating over step sizes dt, gradually decreasing from dt_{max} to dt_{min} by halving the values. The script then saves the solutions of interest in a `csv` file. Another script, `test_methods`, performs the same task and additionally reads the solutions from `ode` and calls the `solutionErrors` function, which evaluates all the relevant errors for various dt and also saves them in a `csv` file. These codes have been valuable as they result in small databases in a single file containing, for all methods and step sizes, the errors that were analyzed. This allows us to recall them later for plotting error trends and evaluating p_{emp} for different dt with `plotErrors` script [5]. We then focused on the relative errors both at the final time and in the L^2 and infinity norms. Regarding the solutions, we experimented with the positions of different particles, noting some variations in the error trends. However, for example, the values of p_{emp} remained roughly stable around certain values for all Cartesian coordinates. We considered then a good idea to concentrate on the error

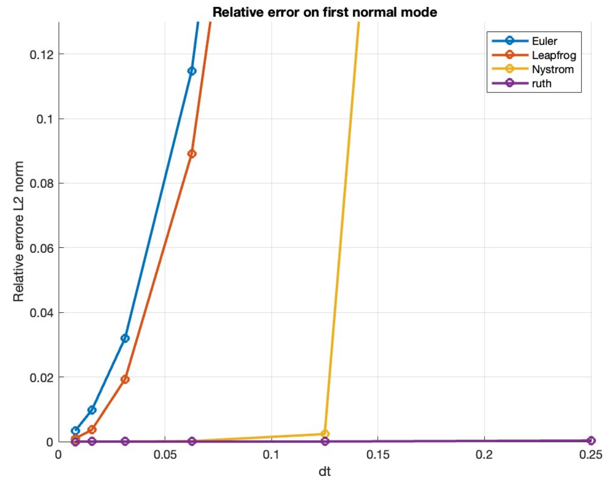

Fig. 6. Relative error in L2 norm on first normal mode for K = 25, we can see how the Ruth method is more stable than others, and gives also smaller error

in normal modes, as in the end, they are the sum of all Cartesian coordinates. For all value of K we considered then the following value of dt: In this way, except for the case

TABLE I
VALUES OF dt HALVING FROM 0.5 TO 0.007812

dt	0.5	0.25	0.125	0.0625	0.03125	0.015625	0.007812

$K = 1$, it was possible to start from outside the stability region of the different methods and then gradually enter it. Thanks to the collected data, for the time steps where the methods were stable, we evaluated both in the L^2 and infinity norms, the p_{emp}, averaging the values once the oscillations had an amplitude smaller than 0.1. The results in the following tables 7, for different values of K, show good agreement with the theoretically expected values, confirming the quality of the implementations, under certain limits, for instance the fact that we are comparing our solutions not with the real exact one. It is important to stress that in our analysis we

	P	P_{emp}(k=1)	P_{emp}(k=25)	P_{emp}(k=100)
Euler	1	1.12	1.22	1.33
Leapfrog	2	1.71	1.60	1.61
Nystrom	3	3.5	3.5	3.5
Ruth	3	3.63	3.54	3.43

Fig. 7. Values of empirical convergence order evaluated on first normal mode with infinity norm error and different value of elastic constant K

keep fixed $t_{\text{max}} = 10^4$, that was a reasonable time for the dynamics observed, but due to limited computational resource we have never reached the limit in which machine error become significant.

[5] In our github page, in the folder `Data` there are some example of data collected and plots

C. Conservation of energy

Starting from the convergence analysis of previous section, the following image 8 shows an interesting evidence for K = 1, namely where all methods are stable in our dt range: This

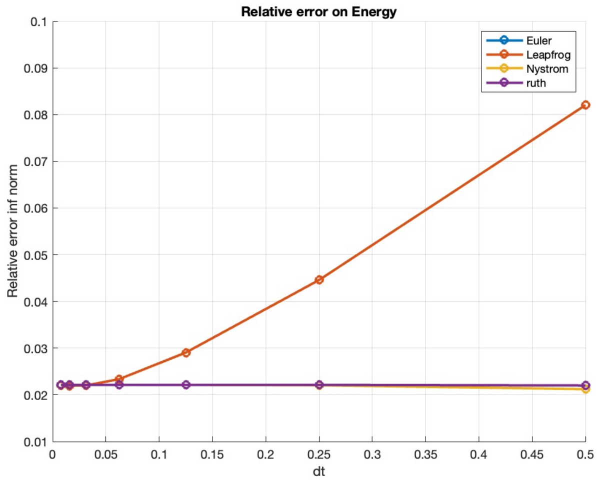

Fig. 8. Relative energy error in infinite norm, please note that Euler is exactly behind Leapfrog and Nystrom under Ruth

is an evidence that, while Euler and Leapfrog has a certain convergence behaviour when $dt \to 0$ despite different from the convergence order of the solution, Nystrom and Ruth show almost a constant low error independently from dt, a notable property in terms of energy accuracy.

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